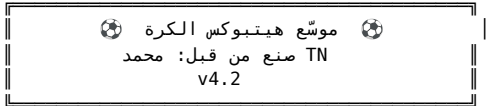


--[[



- (قابل للتغيير) RightShift : إظهار / إخفاء الواجهة
- (قابل للتغيير) H : تشغيل / إيقاف الهيتبوكس
- توقّع مسار الشوت: فيزياء كاملة (جاذبية + قوة الكرة + ارتداد + تعويض البنى)
- يدعم الكمبيوتر والجوال (زر عائم للجوال)
- (writefile يدعم Executor إذا كان الـ) حفظ الإعدادات تلقائياً

]]

-- الخدمات

local Players = game:GetService("Players")
local UserInputService = game:GetService("UserInputService")
local RunService = game:GetService("RunService")
local TweenService = game:GetService("TweenService")
local HttpService = game:GetService("HttpService")
local Workspace = game:GetService("Workspace")

local player = Players.LocalPlayer
local playerGui = player:WaitForChild("PlayerGui")

-- الإعدادات النابتة

local CONFIG = {
 Version = "v4.2",
 Author = "محمد TN",
 SaveFile = "MohammedTN_Hitbox_v4.json",
 MinSize = 0.5,
 MaxSize = 10,
 SizePresets = { 2, 4, 6, 8 },
 AutoSizeRange = 40, -- كل 40 مسافة يتضاعف الحجم في الوضع التلقائي
 AlertSound = "rbxasset://sounds/electronicpingshort.wav",
 Prediction = {
 Step = 1 / 60, -- دقة المحاكاة (60 خطوة بالثانية)
 PointEvery = 3, -- نرسم نقطة كل 3 خطوات
 MaxBounces = 4,
 MaxBalls = 3, -- أقصى عدد كرات نتوقع مسارها بنفس الوقت
 MinSpeed = 4, -- أقل سرعة نعتبر فيها الكرة متحركة
 MaxRange = 350, -- أبعد مسافة للكرة عنك
 SettleSpeed = 2.5, -- تحت هذه السرعة العمودية الكرة تندرج ونوقف
 CalibrationRate = 0.12, -- سرعة المعايرة التلقائية
 CalibrationLimit = 80, -- نتجاهل القفزات الكبيرة (اصطدامات)
 },
 Analysis = {
 KickJump = 18, -- زيادة مفاجئة بالسرعة = ركلة
 MinShotSpeed = 25, -- أقل سرعة نعتبرها شوت
 RecordTime = 3, -- مدة تسجيل المسار الحقيقي بعد الشوت
 MaxOrigins = 20, -- عدد أماكن التسديد المحفوظة
 KickerRange = 10, -- أبعد مسافة بين اللاعب والكرة لحظة الركلة
 PossessionRange = 7, -- الكرة "عند" اللاعب إذا كانت أقرب من كذا
 PossessionSpeed = 15, -- وأبطأ من كذا
 LearnCount = 3, -- نتعلم من آخر 3 شوتات لكل لاعب
 MaxOpponents = 2, -- أقصى عدد خصوم نتوقع شوتهم
 },
 OutlineColors = {
 Color3.fromRGB(124, 92, 255),
 Color3.fromRGB(0, 190, 255),
 Color3.fromRGB(46, 204, 113),
 Color3.fromRGB(255, 200, 0),
 Color3.fromRGB(235, 77, 75),
 Color3.fromRGB(255, 255, 255),
 },
 Themes = {
 { Name = "بنفسجي", Accent = Color3.fromRGB(124, 92, 255), Accent2 = Color3.fromRGB(0, 190, 255) },
 { Name = "أحمر", Accent = Color3.fromRGB(235, 64, 90), Accent2 = Color3.fromRGB(255, 140, 60) },
 { Name = "أخضر", Accent = Color3.fromRGB(46, 204, 113), Accent2 = Color3.fromRGB(0, 200, 180) },
 { Name = "ذهبي", Accent = Color3.fromRGB(255, 190, 40), Accent2 = Color3.fromRGB(255, 120, 40) },
 },
 -- ضع روابطك هنا لتظهر في صفحة "عن السكربت" (اتركها فارغة لإخفائها)
 Links = {
 { Name = "ديسكورد", Url = "" },
 { Name = "يوتيوب", Url = "" },
 },
}

-- الألوان

```
local THEME = {  
    Background = Color3.fromRGB(13, 13, 20),  
    Surface = Color3.fromRGB(21, 21, 32),  
    SurfaceLight = Color3.fromRGB(33, 33, 50),  
    Stroke = Color3.fromRGB(56, 56, 86),  
    Accent = CONFIG.Themes[1].Accent,  
    Accent2 = CONFIG.Themes[1].Accent2,  
    Text = Color3.fromRGB(240, 240, 250),  
    SubText = Color3.fromRGB(148, 148, 176),  
    Success = Color3.fromRGB(46, 204, 113),  
    Danger = Color3.fromRGB(235, 77, 75),  
    Warning = Color3.fromRGB(255, 190, 40),  
}
```

```
local FONT_BOLD = Enum.Font.GothamBold  
local FONT_REG = Enum.Font.Gotham
```

-- تنظيف أي نسخة سابقة من السكريبت

```
if _G.MohammedTN_Hitbox_Cleanup then  
    pcall(_G.MohammedTN_Hitbox_Cleanup)  
end
```

-- الحالة (القيم المحفوظة + قيم التشغيل)

```
local state = {  
    -- محفوظة  
    ballName = "Ball",  
    size = 1,  
    autoSize = false,  
    axisX = 1,  
    axisY = 1,  
    axisZ = 1,  
    showHitbox = true,  
    transparency = 0.5,  
    outline = true,  
    colorIndex = 2,  
    pulse = false,  
    esp = false,  
    tracer = false,  
    prediction = true,  
    predictTime = 2.5,  
    predictBounces = true,  
    predictLag = true,  
    predictCalibrate = true,  
    predictLanding = true,  
    predictClosest = true,  
    predictLabels = true,  
    goalEntry = true,  
    showGoalFrame = true,  
    goals = {},  
    goalWidth = 24,  
    goalHeight = 8,  
    showReplay = true,  
    showOrigins = false,  
    shotAssist = true,  
    assistCamera = false,  
    opponentPredict = true,  
    alert = false,  
    alertDistance = 15,  
    toggleKey = "RightShift",  
    quickKey = "H",  
    themeIndex = 1,  
    floatingButton = UserInputService.TouchEnabled,  
    activeTab = 1,  
  
    -- غير محفوظة  
    enabled = false,  
    guiVisible = true,  
    minimized = false,  
}
```

```
local SAVED_KEYS = {  
    "ballName", "size", "autoSize", "axisX", "axisY", "axisZ", "showHitbox",  
    "transparency", "outline", "colorIndex", "pulse", "esp", "tracer",
```

```

    "prediction", "predictTime", "predictBounces", "predictLag", "predictCalibrate",
    "predictLanding", "predictClosest", "predictLabels", "goalEntry", "showGoalFrame",
    "goals", "goalWidth", "goalHeight", "showReplay", "showOrigins", "shotAssist", "assistCamera", "opponentPredict", "alert", "alertDistance", "toggleKey",
    "quickKey", "themeIndex", "floatingButton", "activeTab",
}

```

```

-----
-- حفظ / تحميل الإعدادات
-----

```

```

local canSave = typeof(writefile) == "function"
    and typeof(readfile) == "function"
    and typeof(isfile) == "function"

local function loadSettings()
    if not canSave then return end
    local ok, data = pcall(function()
        if isfile(CONFIG.SaveFile) then
            return HttpService:JSONDecode(readfile(CONFIG.SaveFile))
        end
        return nil
    end)
    if not ok or type(data) ~= "table" then return end
    for _, key in ipairs(SAVED_KEYS) do
        if data[key] ~= nil and type(data[key]) == type(state[key]) then
            state[key] = data[key]
        end
    end
    -- التحقق من القيم
    local goals = {}
    for _, g in ipairs(state.goals) do
        if type(g) == "table" and #g >= 5 and type(g[1]) == "number" then
            table.insert(goals, g)
        end
    end
    state.goals = goals
    state.colorIndex = math.clamp(math.floor(state.colorIndex), 1, #CONFIG.OutlineColors)
    state.themeIndex = math.clamp(math.floor(state.themeIndex), 1, #CONFIG.Themes)
    if not pcall(function() return Enum.KeyCode[state.toggleKey] end) then state.toggleKey = "RightShift" end
    if not pcall(function() return Enum.KeyCode[state.quickKey] end) then state.quickKey = "H" end
end

```

```

local savePending = false
local function scheduleSave()
    if not canSave or savePending then return end
    savePending = true
    task.delay(0.5, function()
        savePending = false
        local data = {}
        for _, key in ipairs(SAVED_KEYS) do
            data[key] = state[key]
        end
        pcall(function()
            writefile(CONFIG.SaveFile, HttpService:JSONEncode(data))
        end)
    end)
end

```

```

loadSettings()
THEME.Accent = CONFIG.Themes[state.themeIndex].Accent
THEME.Accent2 = CONFIG.Themes[state.themeIndex].Accent2

```

```

-- [part] = { Size, Transparency, CanCollide, Mass, Conn, Box, Esp, EspLabel, Tracer, Calib, LastVel, LastVelTime }
local trackedBalls = {}
local connections = {}

```

```

local function connect(signal, fn)
    local c = signal:Connect(fn)
    table.insert(connections, c)
    return c
end

```

```

-----
-- أدوات مساعدة للواجهة
-----

```

```

local accentRefs = {} -- عناصر تتغير مع الثيم
local gradients = {}

```

```

local function create(className, props, children)
    local inst = Instance.new(className)
    for k, v in pairs(props or {}) do

```

```

        if k ~= "Parent" then
            inst[k] = v
        end
    end
end
for _, child in ipairs(children or {}) do
    child.Parent = inst
end
if props and props.Parent then
    inst.Parent = props.Parent
end
return inst
end

-- ربط خاصية بلون من الثيم ليتحدّث عند تغيير الثيم
local function accentize(inst, prop, key)
    inst[prop] = THEME[key]
    table.insert(accentRefs, { inst, prop, key })
    return inst
end

local function corner(radius)
    return create("UICorner", { CornerRadius = UDim.new(0, radius or 8) })
end

local function stroke(color, thickness, transparency)
    return create("UIStroke", {
        Color = color or THEME.Stroke,
        Thickness = thickness or 1,
        Transparency = transparency or 0,
        ApplyStrokeMode = Enum.ApplyStrokeMode.Border,
    })
end

local function accentGradient(rotation)
    local g = create("UIGradient", {
        Color = ColorSequence.new(THEME.Accent, THEME.Accent2),
        Rotation = rotation or 0,
    })
    table.insert(gradients, g)
    return g
end

local function tween(obj, time, props, style)
    local t = TweenService:Create(
        obj,
        TweenInfo.new(time, style or Enum.EasingStyle.Quint, Enum.EasingDirection.Out),
        props
    )
    t:Play()
    return t
end

local function isPress(input)
    return input.UserInputType == Enum.UserInputType.MouseButton1
        or input.UserInputType == Enum.UserInputType.Touch
end

local function isMove(input)
    return input.UserInputType == Enum.UserInputType.MouseMovement
        or input.UserInputType == Enum.UserInputType.Touch
end

local function hoverButton(btn, normalColor)
    btn.MouseEnter:Connect(function()
        tween(btn, 0.2, { BackgroundColor3 = THEME.Accent, TextColor3 = THEME.Text })
    end)
    btn.MouseLeave:Connect(function()
        tween(btn, 0.2, { BackgroundColor3 = normalColor, TextColor3 = THEME.SubText })
    end)
end

-----
-- الواجهة الرئيسية
-----

local FULL_SIZE = UDim2.new(0, 360, 0, 500)
local MINI_SIZE = UDim2.new(0, 360, 0, 58)

local screenGui = create("ScreenGui", {
    Name = "MohammedTN_HitboxGui",
    ResetOnSpawn = false,
    IgnoreGuiInset = true,

```

```

        ZIndexBehavior = Enum.ZIndexBehavior.Sibling,
        Parent = playerGui,
    })

    -- طبقة الرسم (خطوط التنبّع) خلف الواجهة
    local overlay = create("Frame", {
        Name = "Overlay",
        Size = UDim2.new(1, 0, 1, 0),
        BackgroundTransparency = 1,
        Parent = screenGui,
    })

    local mainFrame = create("Frame", {
        Name = "Main",
        AnchorPoint = Vector2.new(0.5, 0.5),
        Position = UDim2.new(0.5, 0, 0.5, 0),
        Size = FULL_SIZE,
        BackgroundColor3 = THEME.Background,
        BorderSizePixel = 0,
        ClipsDescendants = true,
        Active = true,
        Parent = screenGui,
    }, {
        corner(16),
        stroke(THEME.Stroke, 1.5),
    })

    local mainScale = create("UIScale", { Scale = 0, Parent = mainFrame })

    create("Frame", {
        Size = UDim2.new(1, 0, 0, 3),
        BackgroundColor3 = Color3.new(1, 1, 1),
        BorderSizePixel = 0,
        ZIndex = 5,
        Parent = mainFrame,
    }, { accentGradient(0) })

    -----
    -- شريط العنوان
    -----
    local titleBar = create("Frame", {
        Name = "TitleBar",
        Size = UDim2.new(1, 0, 0, 58),
        BackgroundColor3 = THEME.Surface,
        BorderSizePixel = 0,
        Active = true,
        Parent = mainFrame,
    })

    create("Frame", {
        AnchorPoint = Vector2.new(1, 0.5),
        Position = UDim2.new(1, -14, 0.5, 0),
        Size = UDim2.new(0, 36, 0, 36),
        BackgroundColor3 = Color3.new(1, 1, 1),
        Parent = titleBar,
    }, {
        corner(10),
        accentGradient(45),
        create("TextLabel", {
            Size = UDim2.new(1, 0, 1, 0),
            BackgroundTransparency = 1,
            Text = "⚙",
            TextSize = 20,
            Font = FONT_BOLD,
            TextColor3 = THEME.Text,
        }),
    })

    create("TextLabel", {
        AnchorPoint = Vector2.new(1, 0),
        Position = UDim2.new(1, -60, 0, 10),
        Size = UDim2.new(1, -150, 0, 22),
        BackgroundTransparency = 1,
        Text = "موسّع هينبوكس الكرة",
        TextColor3 = THEME.Text,
        TextSize = 17,
        Font = FONT_BOLD,
        TextXAlignment = Enum.TextXAlignment.Right,
        Parent = titleBar,
    })

```

```

create("TextLabel", {
    AnchorPoint = Vector2.new(1, 0),
    Position = UDim2.new(1, -60, 0, 32),
    Size = UDim2.new(1, -150, 0, 16),
    BackgroundTransparency = 1,
    Text = "صنع من قبل" .. CONFIG.Author .. " • " .. CONFIG.Version,
    TextColor3 = Color3.new(1, 1, 1),
    TextSize = 12,
    Font = FONT_BOLD,
    TextXAlignment = Enum.TextXAlignment.Right,
    Parent = titleBar,
}, { accentGradient(0) })

local function titleButton(text, xOffset, hoverColor)
    local btn = create("TextButton", {
        AnchorPoint = Vector2.new(0, 0.5),
        Position = UDim2.new(0, xOffset, 0.5, 0),
        Size = UDim2.new(0, 30, 0, 30),
        BackgroundColor3 = THEME.SurfaceLight,
        AutoButtonColor = false,
        Text = text,
        TextColor3 = THEME.SubText,
        TextSize = 16,
        Font = FONT_BOLD,
        Parent = titleBar,
    }, { corner(8) })
    btn.MouseEnter:Connect(function()
        tween(btn, 0.2, { BackgroundColor3 = hoverColor or THEME.Accent, TextColor3 = THEME.Text })
    end)
    btn.MouseLeave:Connect(function()
        tween(btn, 0.2, { BackgroundColor3 = THEME.SurfaceLight, TextColor3 = THEME.SubText })
    end)
    return btn
end

local closeBtn = titleButton("x", 14, THEME.Danger)
local minimizeBtn = titleButton("-", 50)

-----
-- التنويبات
-----
local TAB_NAMES = { "إعدادات ⚙️", "أدوات 🛠️", "مظهر 🎨", "تحليل 📊", "توقع 🎯", "هيتبوكس" }
local TAB_COUNT = #TAB_NAMES

local tabBar = create("Frame", {
    Name = "Tabs",
    Position = UDim2.new(0, 12, 0, 66),
    Size = UDim2.new(1, -24, 0, 36),
    BackgroundColor3 = THEME.Surface,
    BorderSizePixel = 0,
    Parent = mainFrame,
}, { corner(10), stroke(THEME.Stroke, 1, 0.5) })

local tabIndicator = create("Frame", {
    Size = UDim2.new(1 / TAB_COUNT, -10, 1, -8),
    Position = UDim2.new(0, 5, 0, 4),
    BackgroundColor3 = Color3.new(1, 1, 1),
    BorderSizePixel = 0,
    Parent = tabBar,
}, { corner(8), accentGradient(0) })

local pageHolder = create("Frame", {
    Name = "Pages",
    Position = UDim2.new(0, 0, 0, 108),
    Size = UDim2.new(1, 0, 1, -108),
    BackgroundTransparency = 1,
    Parent = mainFrame,
})

local pages = {}
local currentPage

-- التنويب الأول على اليمين (ترتيب عربي)
local function tabPosition(index)
    return UDim2.new((TAB_COUNT - index) / TAB_COUNT, 5, 0, 4)
end

local function selectTab(index, animated)
    state.activeTab = index
    for i, page in ipairs(pages) do
        page.Frame.Visible = (i == index)
    end
end

```

```

        tween(page.Button, 0.2, { TextColor3 = (i == index) and THEME.Text or THEME.SubText })
    end
    tween(tabIndicator, animated and 0.3 or 0, { Position = tabPosition(index) })
end

local function newPage(title)
    local index = #pages + 1
    local frame = create("ScrollingFrame", {
        Name = "Page" .. index,
        Size = UDim2.new(1, 0, 1, 0),
        BackgroundTransparency = 1,
        BorderSizePixel = 0,
        CanvasSize = UDim2.new(0, 0, 0, 0),
        AutomaticCanvasSize = Enum.AutomaticSize.Y,
        ScrollBarThickness = 3,
        ScrollingDirection = Enum.ScrollingDirection.Y,
        Visible = false,
        Parent = pageHolder,
    }, {
        create("UIListLayout", {
            Padding = UDim.new(0, 10),
            SortOrder = Enum.SortOrder.LayoutOrder,
        }),
        create("UIPadding", {
            PaddingTop = UDim.new(0, 4),
            PaddingBottom = UDim.new(0, 14),
            PaddingLeft = UDim.new(0, 12),
            PaddingRight = UDim.new(0, 12),
        }),
    })
    accentize(frame, "ScrollBarImageColor3", "Accent")

    local button = create("TextButton", {
        Size = UDim2.new(1 / TAB_COUNT, 0, 1, 0),
        Position = UDim2.new((TAB_COUNT - index) / TAB_COUNT, 0, 0, 0),
        BackgroundTransparency = 1,
        Text = title,
        TextColor3 = THEME.SubText,
        TextScaled = true,
        Font = FONT_BOLD,
        ZIndex = 2,
        Parent = tabBar,
    }, {
        create("UITextSizeConstraint", { MaxTextSize = 12, MinTextSize = 8 }),
        create("UIPadding", { PaddingLeft = UDim.new(0, 3), PaddingRight = UDim.new(0, 3) }),
    })
    button.MouseButton1Click:Connect(function()
        selectTab(index, true)
        scheduleSave()
    end)

    pages[index] = { Frame = frame, Button = button, Order = 0 }
    currentPage = pages[index]
    return currentPage
end

local function nextOrder()
    currentPage.Order += 1
    return currentPage.Order
end

local function card(height)
    return create("Frame", {
        Size = UDim2.new(1, 0, 0, height),
        BackgroundColor3 = THEME.Surface,
        BorderSizePixel = 0,
        LayoutOrder = nextOrder(),
        Parent = currentPage.Frame,
    }, { corner(12), stroke(THEME.Stroke, 1, 0.45) })
end

local function label(parent, props)
    props.BackgroundTransparency = 1
    props.Font = props.Font or FONT_BOLD
    props.TextColor3 = props.TextColor3 or THEME.Text
    props.Parent = parent
    return create("TextLabel", props)
end

local function sectionHeader(text)
    local header = label(currentPage.Frame, {

```

```

        Size = UDim2.new(1, 0, 0, 20),
        Text = text,
        TextSize = 13,
        TextXAlignment = Enum.TextXAlignment.Right,
        LayoutOrder = nextOrder(),
    })
    accentize(header, "TextColor3", "Accent2")
end

local function cardHeading(parent, text, y)
    return label(parent, {
        AnchorPoint = Vector2.new(1, 0),
        Position = UDim2.new(1, -14, 0, y or 10),
        Size = UDim2.new(1, -150, 0, 20),
        Text = text,
        TextSize = 15,
        TextXAlignment = Enum.TextXAlignment.Right,
    })
end

local function cardTitle(parent, title, subtitle)
    label(parent, {
        AnchorPoint = Vector2.new(1, 0),
        Position = UDim2.new(1, -14, 0, 9),
        Size = UDim2.new(1, -90, 0, 18),
        Text = title,
        TextSize = 15,
        TextXAlignment = Enum.TextXAlignment.Right,
    })
    return label(parent, {
        AnchorPoint = Vector2.new(1, 0),
        Position = UDim2.new(1, -14, 0, 28),
        Size = UDim2.new(1, -90, 0, 14),
        Text = subtitle,
        TextColor3 = THEME.SubText,
        TextSize = 12,
        Font = FONT_REG,
        TextXAlignment = Enum.TextXAlignment.Right,
    })
end

-- مفتاح تشغيل/إيقاف متحرك
local function makeSwitch(parent, initial, onChanged)
    local switch = create("TextButton", {
        AnchorPoint = Vector2.new(0, 0.5),
        Position = UDim2.new(0, 14, 0.5, 0),
        Size = UDim2.new(0, 48, 0, 26),
        BackgroundColor3 = THEME.SurfaceLight,
        AutoButtonColor = false,
        Text = "",
        Parent = parent,
    }, { corner(13) })

    local gradient = accentGradient(0)
    gradient.Parent = switch

    local knob = create("Frame", {
        AnchorPoint = Vector2.new(0, 0.5),
        Position = UDim2.new(0, 3, 0.5, 0),
        Size = UDim2.new(0, 20, 0, 20),
        BackgroundColor3 = THEME.SubText,
        Parent = switch,
    }, { corner(10) })

    local value = initial
    local api = {}

    local function render(animated)
        gradient.Enabled = value
        switch.BackgroundColor3 = value and Color3.new(1, 1, 1) or THEME.SurfaceLight
        tween(knob, animated and 0.25 or 0, {
            Position = value and UDim2.new(1, -23, 0.5, 0) or UDim2.new(0, 3, 0.5, 0),
            BackgroundColor3 = value and Color3.new(1, 1, 1) or THEME.SubText,
        })
    end

    function api.set(newValue, silent)
        if newValue == value then return end
        value = newValue
        render(true)
        if not silent then

```



```

        onChanged(value)
    end
end

switch.MouseButton1Click:Connect(function()
    api.set(not value)
end)

render(false)
return api
end

-- بطاقة فيها عنوان + مفتاح مربوط بقيمة في
local function switchCard(title, subtitle, key, onExtra)
    local c = card(54)
    local sub = cardTitle(c, title, subtitle)
    local sw = makeSwitch(c, state[key], function(value)
        state[key] = value
        scheduleSave()
        if onExtra then
            onExtra(value)
        end
    end)
    return sw, sub, c
end

-- شريط تمرير قابل لإعادة الاستخدام (الرقم يظهر فوقه على اليسار)
local activeSlider = nil

local function makeSlider(parent, y, min, max, step, initial, format, onChanged)
    local page = currentPage.Frame

    local bg = create("Frame", {
        Position = UDim2.new(0, 14, 0, y),
        Size = UDim2.new(1, -28, 0, 10),
        BackgroundColor3 = THEME.SurfaceLight,
        Active = true,
        Parent = parent,
    }, { corner(5) })

    local fill = create("Frame", {
        Size = UDim2.new(0, 0, 1, 0),
        BackgroundColor3 = Color3.new(1, 1, 1),
        BorderSizePixel = 0,
        Parent = bg,
    }, { corner(5), accentGradient(0) })

    local knobStroke = stroke(THEME.Accent, 2)
    accentize(knobStroke, "Color", "Accent")
    local knob = create("Frame", {
        AnchorPoint = Vector2.new(0.5, 0.5),
        Position = UDim2.new(0, 0, 0.5, 0),
        Size = UDim2.new(0, 18, 0, 18),
        BackgroundColor3 = Color3.new(1, 1, 1),
        ZIndex = 2,
        Parent = bg,
    }, { corner(9), knobStroke })

    local badge = create("TextLabel", {
        Position = UDim2.new(0, 14, 0, y - 38),
        Size = UDim2.new(0, 60, 0, 24),
        BackgroundColor3 = THEME.SurfaceLight,
        Text = "",
        TextSize = 14,
        Font = FONT_BOLD,
        Parent = parent,
    }, { corner(6) })
    accentize(badge, "TextColor3", "Accent2")

    local api = { value = initial }

    function api.set(value, animated, silent)
        value = math.clamp(math.floor(value / step + 0.5) * step, min, max)
        api.value = value
        local percent = (value - min) / (max - min)
        local t = animated and 0.25 or 0
        tween(fill, t, { Size = UDim2.new(percent, 0, 1, 0) })
        tween(knob, t, { Position = UDim2.new(percent, 0, 0.5, 0) })
        badge.Text = string.format(format, value)
        if not silent then
            onChanged(value)
        end
    end
end

```

```

        end
    end

    function api.update(x)
        local percent = math.clamp((x - bg.AbsolutePosition.X) / bg.AbsoluteSize.X, 0, 1)
        api.set(min + percent * (max - min), false)
    end

    function api.release()
        page.ScrollingEnabled = true
        tween(knob, 0.15, { Size = UDim2.new(0, 18, 0, 18) })
    end

    bg.InputBegan:Connect(function(input)
        if isPress(input) then
            activeSlider = api
            page.ScrollingEnabled = false
            tween(knob, 0.15, { Size = UDim2.new(0, 22, 0, 22) })
            api.update(input.Position.X)
        end
    end)

    api.set(initial, false, true)
    return api
end

-- بطاقة state فيها عنوان + شريط مربوط بقيمة في
local function sliderCard(title, key, min, max, step, format)
    local c = card(70)
    cardHeading(c, title)
    return makeSlider(c, 46, min, max, step, state[key], format, function(value)
        state[key] = value
        scheduleSave()
    end)
end

local function smallButton(parent, text, position, size)
    local btn = create("TextButton", {
        Position = position,
        Size = size,
        BackgroundColor3 = THEME.SurfaceLight,
        AutoButtonColor = false,
        Text = text,
        TextColor3 = THEME.SubText,
        TextSize = 12,
        Font = FONT_BOLD,
        Parent = parent,
    }, { corner(6) })
    hoverButton(btn, THEME.SurfaceLight)
    return btn
end

local function horizontalRow(parent, y, height, alignment)
    return create("Frame", {
        Position = UDim2.new(0, 14, 0, y),
        Size = UDim2.new(1, -28, 0, height),
        BackgroundTransparency = 1,
        Parent = parent,
    }, {
        create("UIListLayout", {
            FillDirection = Enum.FillDirection.Horizontal,
            HorizontalAlignment = alignment or Enum.HorizontalAlignment.Left,
            Padding = UDim.new(0, 8),
            SortOrder = Enum.SortOrder.LayoutOrder,
        })
    })
end

-- سطر معلومات (اسم على اليمين وقيمة على اليسار)
local function statRow(parent, y, name)
    label(parent, {
        AnchorPoint = Vector2.new(1, 0),
        Position = UDim2.new(1, -14, 0, y),
        Size = UDim2.new(0.55, -14, 0, 18),
        Text = name,
        TextColor3 = THEME.SubText,
        TextSize = 13,
        Font = FONT_REG,
        TextXAlignment = Enum.TextXAlignment.Right,
    })
    return label(parent, {

```

```

        Position = UDim2.new(0, 14, 0, y),
        Size = UDim2.new(0.45, -14, 0, 18),
        Text = "-",
        TextSize = 13,
        TextXAlignment = Enum.TextXAlignment.Left,
    })
end

-----
-- الإشعارات
-----

local toastHolder = create("Frame", {
    AnchorPoint = Vector2.new(1, 1),
    Position = UDim2.new(1, -16, 1, -16),
    Size = UDim2.new(0, 260, 1, -32),
    BackgroundTransparency = 1,
    ZIndex = 10,
    Parent = screenGui,
}, {
    create("UIListLayout", {
        Padding = UDim.new(0, 8),
        VerticalAlignment = Enum.VerticalAlignment.Bottom,
        HorizontalAlignment = Enum.HorizontalAlignment.Right,
        SortOrder = Enum.SortOrder.LayoutOrder,
    }),
})

local toastOrder = 0
local function notify(text, color)
    toastOrder += 1
    local toast = create("Frame", {
        Size = UDim2.new(1, 0, 0, 46),
        BackgroundColor3 = THEME.Surface,
        BackgroundTransparency = 1,
        LayoutOrder = toastOrder,
        Parent = toastHolder,
    }, { corner(10) })

    local toastStroke = stroke(THEME.Stroke, 1, 1)
    toastStroke.Parent = toast

    local bar = create("Frame", {
        AnchorPoint = Vector2.new(1, 0),
        Position = UDim2.new(1, 0, 0, 0),
        Size = UDim2.new(0, 4, 1, 0),
        BackgroundColor3 = color or THEME.Accent,
        BackgroundTransparency = 1,
        BorderSizePixel = 0,
        Parent = toast,
    }, { corner(2) })

    local toastLabel = label(toast, {
        Position = UDim2.new(0, 12, 0, 0),
        Size = UDim2.new(1, -28, 1, 0),
        Text = text,
        TextTransparency = 1,
        TextSize = 13,
        TextWrapped = true,
        TextXAlignment = Enum.TextXAlignment.Right,
    })

    tween(toast, 0.3, { BackgroundTransparency = 0.05 })
    tween(toastStroke, 0.3, { Transparency = 0.3 })
    tween(bar, 0.3, { BackgroundTransparency = 0 })
    tween(toastLabel, 0.3, { TextTransparency = 0 })

    task.delay(2.5, function()
        if not toast.Parent then return end
        tween(toast, 0.3, { BackgroundTransparency = 1 })
        tween(toastStroke, 0.3, { Transparency = 1 })
        tween(bar, 0.3, { BackgroundTransparency = 1 })
        tween(toastLabel, 0.3, { TextTransparency = 1 }).Completed:Wait()
        toast:Destroy()
    end)
end

local alertSound = create("Sound", {
    SoundId = CONFIG.AlertSound,
    Volume = 1,
    Parent = screenGui,
})

```

```

-----
-- منطق الكرة
-----
تُنشأ لاحقاً في بطاقة المعلومات --
local ballCountLabel, nearestLabel

local function outlineColor()
    return CONFIG.OutlineColors[state.colorIndex]
end

local function countBalls()
    local n = 0
    for _ in pairs(trackedBalls) do
        n += 1
    end
    return n
end

local function refreshBallCount()
    if ballCountLabel then
        ballCountLabel.Text = tostring(countBalls())
    end
end

local function restoreBall(part, props)
    if part.Parent then
        part.Size = props.Size
        part.Transparency = props.Transparency
        part.CanCollide = props.CanCollide
    end
end

local function restoreAll()
    for part, props in pairs(trackedBalls) do
        restoreBall(part, props)
    end
end

local function untrackBall(part)
    local props = trackedBalls[part]
    if not props then return end
    restoreBall(part, props)
    props.Conn:Disconnect()
    props.Box:Destroy()
    props.Esp:Destroy()
    props.Tracer:Destroy()
    trackedBalls[part] = nil
    refreshBallCount()
end

local function addBall(part)
    if not part:IsA("BasePart") or trackedBalls[part] then return end

    local props = {
        Size = part.Size,
        Transparency = part.Transparency,
        CanCollide = part.CanCollide,
        -- الكتلة الأصلية (قبل تكبير الهيتيوكس) لحساب قوة الكرة بدقة
        Mass = math.max(part.AssemblyMass, 0.01),
        Calib = Vector3.zero,
        LastVel = part.AssemblyLinearVelocity,
        LastVelTime = os.clock(),
    }

    -- إطار يوضّح حدود الهيتيوكس
    props.Box = create("SelectionBox", {
        Adornee = part,
        Color3 = outlineColor(),
        LineThickness = 0.05,
        SurfaceTransparency = 1,
        Visible = false,
        Parent = playerGui,
    })

    -- مؤشر فوق الكرة: المسافة + السرعة
    props.Esp = create("BillboardGui", {
        Adornee = part,
        AlwaysOnTop = true,
        Size = UDim2.new(0, 150, 0, 26),
        StudsOffset = Vector3.new(0, 3, 0),
        Enabled = false,
    })

```

```

        Parent = playerGui,
    })
    local espBg = create("Frame", {
        Size = UDim2.new(1, 0, 1, 0),
        BackgroundColor3 = THEME.Background,
        BackgroundTransparency = 0.25,
        Parent = props.Esp,
    }, { corner(8) })
    local espStroke = stroke(THEME.Accent, 1.5)
    espStroke.Parent = espBg
    accentize(espStroke, "Color", "Accent")
    props.EspLabel = label(props.Esp, {
        Size = UDim2.new(1, 0, 1, 0),
        Text = "🏀",
        TextSize = 13,
        ZIndex = 2,
    })

    -- خط التتبع
    props.Tracer = create("Frame", {
        AnchorPoint = Vector2.new(0.5, 0.5),
        BackgroundColor3 = outlineColor(),
        BorderSizePixel = 0,
        Visible = false,
        Parent = overlay,
    })

    props.Conn = part.AncestryChanged:Connect(function(_, parent)
        if parent == nil then
            untrackBall(part)
        end
    end)

    trackedBalls[part] = props
    refreshBallCount()
end

local function scanForBalls()
    for _, descendant in ipairs(Workspace:GetDescendants()) do
        if descendant.Name == state.ballName then
            addBall(descendant)
        end
    end
end

local function rescan()
    for part in pairs(trackedBalls) do
        untrackBall(part)
    end
    scanForBalls()
    refreshBallCount()
end

connect(Workspace.DescendantAdded, function(descendant)
    if descendant.Name == state.ballName then
        addBall(descendant)
    end
end)

local function getRootPosition()
    local character = player.Character
    local root = character and character:FindFirstChild("HumanoidRootPart")
    return root and root.Position
end

local function drawLine(frame, a, b)
    local delta = b - a
    frame.Size = UDim2.new(0, delta.Magnitude, 0, 2)
    frame.Position = UDim2.new(0, (a.X + b.X) / 2, 0, (a.Y + b.Y) / 2)
    frame.Rotation = math.deg(math.atan2(delta.Y, delta.X))
end

-----
-- توقع مسار الشوت (محاكاة فيزيائية) --
-----

local vizFolder = create("Folder", {
    Name = "MohammedTN_Prediction",
    Parent = Workspace.CurrentCamera,
})

local rayParams = RaycastParams.new()

```

```
rayParams.FilterType = Enum.RaycastFilterType.Exclude
rayParams.IgnoreWater = true
```

-- نتجاهل الكرات واللاعبين ورسومنا عند فحص الاصطدام

```
local function refreshRayFilter()
    local list = { vizFolder }
    for part in pairs(trackedBalls) do
        table.insert(list, part)
    end
    for _, name in ipairs({ "Footballs", "Characters" }) do
        local folder = Workspace:FindFirstChild(name)
        if folder then
            table.insert(list, folder)
        end
    end
    for _, plr in ipairs(Players:GetPlayers()) do
        if plr.Character then
            table.insert(list, plr.Character)
        end
    end
    rayParams.FilterDescendantsInstances = list
end
```

-- مقسومة على كتلتها الأصلية (BodyForce / VectorForce) قوة الكرة

```
local function ballForceAcceleration(part, props)
    local total = Vector3.zero
    for _, child in ipairs(part:GetChildren()) do
        if child:IsA("BodyForce") then
            total += child.Force
        elseif child:IsA("VectorForce") and child.Enabled then
            local force = child.Force
            if child.RelativeTo == Enum.ActuatorRelativeTo.Attachment0 and child.Attachment0 then
                force = child.Attachment0.WorldCFrame:VectorToWorldSpace(force)
            end
            total += force
        end
    end
    return total / props.Mass
end
```

```
local function modelAcceleration(part, props)
    return Vector3.new(0, -Workspace.Gravity, 0) + ballForceAcceleration(part, props)
end
```

-- معايرة تلقائية: نقارن التسارع الحقيقي بالمحسوب ونصح الفرق

```
local function calibrate(part, props, now)
    local velocity = part.AssemblyLinearVelocity
    if velocity == props.LastVel then return end -- ما وصل تحديث جديد من السيرفر
    local dt = now - props.LastVelTime
    if dt > 0.005 and dt < 0.25 then
        local measured = (velocity - props.LastVel) / dt
        local residual = measured - modelAcceleration(part, props)
        local P = CONFIG.Prediction
        if residual.Magnitude < P.CalibrationLimit then
            props.Calib = props.Calib:Lerp(residual, P.CalibrationRate)
        else
            -- اصطدام أو ركلة: نخفف المعايرة بدل ما نخربها
            props.Calib *= 0.5
        end
    end
    props.LastVel = velocity
    props.LastVelTime = now
end
```

```
local function networkLag()
    local ok, ping = pcall(function()
        return player:GetNetworkPing()
    end)
    return ok and math.clamp(ping, 0, 0.5) or 0
end
```

-- يحاكي حركة الكرة ويرجع نقاط المسار والأحداث المهمة

-- options: { NoLag = true, Duration = ثواني } (للمقارنة مع المسار الحقيقي)

```
local function simulate(part, props, rootPos, options)
    options = options or {}
    local P = CONFIG.Prediction
    local step = P.Step
    local duration = options.Duration or state.predictTime
    local radius = math.min(props.Size.X, props.Size.Y, props.Size.Z) / 2
    local accel = modelAcceleration(part, props)
    if state.predictCalibrate then
```

```

    accel += props.Calib
end

-- محاكي شوت افتراضي (مساعد التسديد)
local pos = options.Position or part.Position
local vel = options.Velocity or part.AssemblyLinearVelocity

-- تعويض البنى: الكرة الحقيقية متقدمة عن اللي نشوفه
if state.predictLag and not options.NoLag and not options.Velocity then
    local lag = networkLag()
    pos += vel * lag + 0.5 * accel * lag * lag
    vel += accel * lag
end

-- الكرة على الأرض وتندرج: نلعي الجاذبية باتجاه الأرض ونرفعها شوي
-- (فحص الاصطدام يتجاهل السطح اللي الكرة لامسته من البداية)
local ground = Workspace.Raycast(pos, Vector3.new(0, -(radius + 0.35), 0), rayParams)
if ground and ground.Normal.Y > 0.6 and vel:Dot(ground.Normal) < P.SettleSpeed then
    local normal = ground.Normal
    accel -= normal * accel:Dot(normal)
    vel -= normal * math.min(vel:Dot(normal), 0)
    pos = ground.Position + normal * (radius + 0.05)
end

local physical = part.CurrentPhysicalProperties
local elasticity = physical and physical.Elasticity or 0.5
local friction = physical and physical.Friction or 0.3

local result = {
    Points = { pos },
    Times = { 0 }, -- وقت كل نقطة (لحساب دخول المرمى ومقارنة الدقة)
    Radius = radius,
    Bounces = {},
    Landing = nil, -- { Position, Time, Normal }
    Closest = nil, -- { Position, Time, Distance }
    Start = pos,
}

local t = 0
local stepIndex = 0
local bounces = 0

local function checkClosest(point, time)
    if not rootPos then return end
    local distance = (point - rootPos).Magnitude
    if not result.Closest or distance < result.Closest.Distance then
        result.Closest = { Position = point, Time = time, Distance = distance }
    end
end

checkClosest(pos, 0)

local function addPoint(point, time)
    table.insert(result.Points, point)
    table.insert(result.Times, time)
end

while t < duration do
    local nextVel = vel + accel * step
    local move = (vel + nextVel) * 0.5 * step
    local hit = move.Magnitude > 1e-4 and Workspace.Spherecast(pos, radius, move, rayParams)

    if hit then
        local fraction = math.clamp(hit.Distance / move.Magnitude, 0, 1)
        pos += move.Unit * hit.Distance
        vel += accel * step * fraction
        t += step * fraction
        addPoint(pos, t)
        checkClosest(pos, t)

        local normal = hit.Normal
        if not result.Landing and normal.Y > 0.6 then
            result.Landing = { Position = pos - normal * radius, Time = t, Normal = normal }
        end
        table.insert(result.Bounces, pos - normal * radius)

        if not state.predictBounces or bounces >= P.MaxBounces then
            break
        end
    end

    -- ارتداد: نعكس السرعة العمودية ونخفف الأفقية بالاحتكاك

```

```

        local vn = vel:Dot(normal)
        if vn < 0 then
            local normalVel = normal * vn
            local tangentVel = vel - normalVel
            vel = tangentVel * (1 - math.clamp(friction, 0, 1) * 0.25) - normalVel * elasticity
        end
        if normal.Y > 0.6 and math.abs(vel:Dot(normal)) < P.SettleSpeed then
            -- الكرة صارت تندرج: نكمل الحساب على الأرض (الشوات الأرضية)
            vel -= normal * vel:Dot(normal)
            accel -= normal * accel:Dot(normal)
            pos += normal * 0.05
            if vel.Magnitude < 1 then
                break -- الكرة وقفت
            end
        else
            pos += normal * 0.02
        end
        bounces += 1
    else
        pos += move
        vel = nextVel
        t += step
        stepIndex += 1
        if stepIndex % P.PointEvery == 0 then
            addPoint(pos, t)
        end
        checkClosest(pos, t)
    end
end

addPoint(pos, t)
return result
end

```

```

-- مخزن قطع الرسم (نعيد استخدامها بدل ما ننشئ قطع كل فريم)
local pool = { Segment = {}, Bounce = {}, Landing = {}, Closest = {}, Goal = {}, Origin = {} }
local used = { Segment = 0, Bounce = 0, Landing = 0, Closest = 0, Goal = 0, Origin = 0 }

```

```

local function vizPart(shape, size)
    return create("Part", {
        Anchored = true,
        CanCollide = false,
        CanQuery = false,
        CanTouch = false,
        CastShadow = false,
        Locked = true,
        Material = Enum.Material.Neon,
        Shape = shape,
        Size = size,
        Transparency = 1,
        Parent = vizFolder,
    })
end

```

```

local function vizLabel(adornee, offset)
    local billboard = create("BillboardGui", {
        Adornee = adornee,
        -- ما نرسمها فوق كل شي عشان ما تغطي الكرة
        AlwaysOnTop = false,
        LightInfluence = 0,
        MaxDistance = 220,
        Size = UDim2.new(0, 96, 0, 20),
        StudsOffset = offset,
        Enabled = false,
        Parent = vizFolder,
    })
    create("Frame", {
        Size = UDim2.new(1, 0, 1, 0),
        BackgroundColor3 = THEME.Background,
        BackgroundTransparency = 0.35,
        Parent = billboard,
    }, { corner(6) })
    local text = label(billboard, {
        Size = UDim2.new(1, 0, 1, 0),
        Text = "",
        TextSize = 11,
        ZIndex = 2,
    })
    return billboard, text
end

```



```

local factories = {
    Segment = function()
        return { Part = vizPart(Enum.PartType.Block, Vector3.new(0.18, 0.18, 1)) }
    end,
    Bounce = function()
        return { Part = vizPart(Enum.PartType.Ball, Vector3.new(0.6, 0.6, 0.6)) }
    end,
    Landing = function()
        local part = vizPart(Enum.PartType.Cylinder, Vector3.new(0.15, 4, 4))
        local billboard, text = vizLabel(part, Vector3.new(0, 2.5, 0))
        return { Part = part, Billboard = billboard, Text = text }
    end,
    Closest = function()
        local part = vizPart(Enum.PartType.Ball, Vector3.new(1.1, 1.1, 1.1))
        local billboard, text = vizLabel(part, Vector3.new(0, 2, 0))
        return { Part = part, Billboard = billboard, Text = text }
    end,
    Goal = function()
        local part = vizPart(Enum.PartType.Block, Vector3.new(2.4, 2.4, 0.15))
        local billboard, text = vizLabel(part, Vector3.new(0, 2.4, 0))
        return { Part = part, Billboard = billboard, Text = text }
    end,
    Origin = function()
        return { Part = vizPart(Enum.PartType.Cylinder, Vector3.new(0.12, 1.6, 1.6)) }
    end,
}

local function take(kind)
    used[kind] += 1
    local item = pool[kind][used[kind]]
    if not item then
        item = factories[kind]()
        pool[kind][used[kind]] = item
    end
    return item
end

local function hideUnused()
    for kind, list in pairs(pool) do
        for i = used[kind] + 1, #list do
            local item = list[i]
            item.Part.Transparency = 1
            if item.Billboard then
                item.Billboard.Enabled = false
            end
        end
        used[kind] = 0
    end
end

local function drawSegment(a, b, color, transparency, thickness)
    local length = (b - a).Magnitude
    if length < 0.05 then return end
    thickness = thickness or 0.18
    local seg = take("Segment").Part
    seg.Size = Vector3.new(thickness, thickness, length)
    seg.CFrame = CFrame.lookAt((a + b) / 2, b)
    seg.Color = color
    seg.Transparency = transparency
end

local function drawPrediction(result, color, showLabels, ballPos, clearRadius)
    local points = result.Points
    local count = #points
    for i = 1, count - 1 do
        local a, b = points[i], points[i + 1]
        -- ما نرسم داخل الكرة نفسها عشان تبان واضحة
        local insideBall = (a - ballPos).Magnitude < clearRadius and (b - ballPos).Magnitude < clearRadius
        if not insideBall then
            -- يهت الخط كل ما بعد في المستقبل
            drawSegment(a, b, color, 0.1 + 0.6 * (i / count))
        end
    end

    if state.predictBounces then
        for _, point in ipairs(result.Bounces) do
            local bounce = take("Bounce").Part
            bounce.CFrame = CFrame.new(point)
            bounce.Color = color
            bounce.Transparency = 0.25
        end
    end
end

```

```

end

if state.predictLanding and result.Landing then
    local marker = take("Landing")
    local landing = result.Landing
    marker.Part.CFrame = CFrame.new(landing.Position + landing.Normal * 0.08) * CFrame.Angles(0, 0, math.pi / 2)
    marker.Part.Color = color
    marker.Part.Transparency = 0.35
    marker.Text.Text = string.format("⌚ %.2f ث", landing.Time)
    marker.Text.TextColor3 = color
    marker.Billboard.Enabled = showLabels
end

local closest = result.Closest
if state.predictClosest and closest and closest.Time > 0.05
    and (closest.Position - ballPos).Magnitude > clearRadius
then
    local marker = take("Closest")
    marker.Part.CFrame = CFrame.new(closest.Position)
    marker.Part.Color = THEME.Warning
    marker.Part.Transparency = 0.3
    marker.Text.Text = string.format("⚡ %.1f 2.% * م ث", closest.Distance, closest.Time)
    marker.Text.TextColor3 = THEME.Warning
    marker.Billboard.Enabled = showLabels
end
end

-----
-- الحلقة الرئيسية
-----
local predictionLabels = {} -- تُنشأ في صفحة التوقع
local statLabels = {}      -- تُنشأ في صفحة التحليل

-----
-- التحليل: المرمى + الإحصائيات + آخر شوت
-----
local UP = Vector3.new(0, 1, 0)
local goalCache = nil

-- نقطة نص خط المرمى + اتجاه الملعب { x, y, z, nx, nz }: المرمى محفوظ كـ
local function getGoals()
    if goalCache then return goalCache end
    goalCache = {}
    for _, g in ipairs(state.goals) do
        if type(g) == "table" and #g >= 5 then
            local normal = Vector3.new(g[4], 0, g[5])
            if normal.Magnitude > 0.1 then
                normal = normal.Unit
                table.insert(goalCache, {
                    Origin = Vector3.new(g[1], g[2], g[3]),
                    Normal = normal,
                    Right = UP:Cross(normal),
                })
            end
        end
    end
    return goalCache
end

-- أول مرة يقطع فيها المسار خط المرمى (من جهة الملعب)
local function findGoalCrossing(result)
    local points, times = result.Points, result.Times
    local best
    for _, goal in ipairs(getGoals()) do
        for i = 1, #points - 1 do
            local a, b = points[i], points[i + 1]
            local da = (a - goal.Origin):Dot(goal.Normal)
            local db = (b - goal.Origin):Dot(goal.Normal)
            if da > 0 and db <= 0 then
                local alpha = da / (da - db)
                local point = a:Lerp(b, alpha)
                local time = times[i] + (times[i + 1] - times[i]) * alpha
                local lateral = (point - goal.Origin):Dot(goal.Right)
                local height = point.Y - goal.Origin.Y
                -- نعرض الشوتات التي تمر قريب من المرمى بس
                if math.abs(lateral) <= state.goalWidth and height <= state.goalHeight * 2 then
                    local inside = math.abs(lateral) <= state.goalWidth / 2 + result.Radius
                    and height >= -result.Radius
                    and height <= state.goalHeight + result.Radius
                    if not best or time < best.Time then
                        best = { Position = point, Time = time, Inside = inside, Goal = goal }
                    end
                end
            end
        end
    end
end

```

```

        end
    end
    break
end
end
end
return best
end

local function drawGoalCrossing(crossing, showLabels)
    local marker = take("Goal")
    local color = crossing.Inside and THEME.Danger or THEME.Warning
    marker.Part.CFrame = CFrame.lookAt(crossing.Position, crossing.Position + crossing.Goal.Normal)
    marker.Part.Color = color
    marker.Part.Transparency = 0.3
    marker.Text.Text = string.format(crossing.Inside and "🚩 2.% • ث هدفى" or "2.% • ث فبّرّا", crossing.Time)
    marker.Text.TextColor3 = color
    marker.Billboard.Enabled = showLabels
end

local function drawGoalFrames()
    local white = Color3.new(1, 1, 1)
    for _, goal in ipairs(getGoals()) do
        local half = goal.Right * (state.goalWidth / 2)
        local top = UP * state.goalHeight
        local left, right = goal.Origin - half, goal.Origin + half
        drawSegment(left, left + top, white, 0.45, 0.25)
        drawSegment(right, right + top, white, 0.45, 0.25)
        drawSegment(left + top, right + top, white, 0.45, 0.25)
        drawSegment(left, right, white, 0.6, 0.15)
    end
end

-- إحصائيات الشوتات |
local stats = { Shots = 0, MaxSpeed = 0, TotalSpeed = 0, LastSpeed = 0, LastError = nil }
local shotOrigins = {}
local lastShot = nil -- { Ball, StartTime, Predicted, Actual, ActualTimes, LastSample, Done }

local function refreshStats()
    if not statLabels.Shots then return end
    statLabels.Shots.Text = tostring(stats.Shots)
    statLabels.Max.Text = stats.Shots > 0 and string.format("%.1f", stats.MaxSpeed) or "-"
    statLabels.Avg.Text = stats.Shots > 0 and string.format("%.1f", stats.TotalSpeed / stats.Shots) or "-"
    statLabels.Last.Text = stats.Shots > 0 and string.format("%.1f", stats.LastSpeed) or "-"
    statLabels.Error.Text = stats.LastError and string.format("%.2f م", stats.LastError) or "-"
end

-- مكان الكرة المتوقع في وقت معيّن
local function pointAtTime(points, times, t)
    for i = 2, #points do
        if times[i] >= t then
            local t0, t1 = times[i - 1], times[i]
            local alpha = t1 > t0 and (t - t0) / (t1 - t0) or 0
            return points[i - 1]:Lerp(points[i], alpha)
        end
    end
    return nil
end

-- نقارن المسار الحقيقي بالمتوقع ونحسب متوسط الخطأ
local function finishShot()
    if not lastShot or lastShot.Done then return end
    lastShot.Done = true
    local predicted = lastShot.Predicted
    local total, count = 0, 0
    for i = 2, #lastShot.Actual do
        local expected = pointAtTime(predicted.Points, predicted.Times, lastShot.ActualTimes[i])
        if expected then
            total += (expected - lastShot.Actual[i]).Magnitude
            count += 1
        end
    end
    if count > 0 then
        stats.LastError = total / count
        refreshStats()
    end
end

-- تعلّم أسلوب الشوت: نحفظ سرعة الكرة بالنسبة لاتجاه اللاعب (قدام / فوق / يمين)
local kickHistory = {} -- {UserId} = { {F, U, R}, ... }
local globalHistory = {}

```

```

local function flatForward(vector)
    local flat = Vector3.new(vector.X, 0, vector.Z)
    return flat.Magnitude > 0.05 and flat.Unit or nil
end

local function toLocalShot(forward, velocity)
    local right = forward:Cross(UP)
    return { F = velocity:Dot(forward), U = velocity.Y, R = velocity:Dot(right) }
end

local function fromLocalShot(forward, shot)
    local right = forward:Cross(UP)
    return forward * shot.F + UP * shot.U + right * shot.R
end

local function averageShot(list, count)
    local n = math.min(#list, count)
    if n == 0 then return nil end
    local f, u, r = 0, 0, 0
    for i = #list - n + 1, #list do
        f += list[i].F
        u += list[i].U
        r += list[i].R
    end
    return { F = f / n, U = u / n, R = r / n }
end

local function learnedShot(userId)
    local own = kickHistory[userId]
    return (own and averageShot(own, CONFIG.Analysis.LearnCount))
        or averageShot(globalHistory, 10)
end

local function rootOf(plr)
    local character = plr.Character
    return character and character:FindFirstChild("HumanoidRootPart")
end

-- أقرب لاعب للكرة لحظة الركلة = اللي سددها
local function findKicker(position)
    local best, bestRoot, bestDistance = nil, nil, CONFIG.Analysis.KickerRange
    for _, plr in ipairs(Players:GetPlayers()) do
        local root = rootOf(plr)
        if root then
            local distance = (root.Position - position).Magnitude
            if distance < bestDistance then
                best, bestRoot, bestDistance = plr, root, distance
            end
        end
    end
    return best, bestRoot
end

local learnLabel -- بُنِشْأ في صفحة التحليل

local function refreshLearnLabel()
    if not learnLabel then return end
    local own = kickHistory[player.UserId]
    learnLabel.Text = string.format("%d شوتات • %d كل الشوتات", own and #own or 0, #globalHistory)
end

local function learnKick(part, velocity)
    -- الكرة تحركت شوي بعد الركلة، فنرجع لمكانها قبل فريم تقريباً
    local kicker, root = findKicker(part.Position - velocity / 60)
    if not kicker then return end
    local forward = flatForward(root.CFrame.LookVector)
    if not forward then return end
    local shot = toLocalShot(forward, velocity)
    local list = kickHistory[kicker.UserId] or {}
    kickHistory[kicker.UserId] = list
    table.insert(list, shot)
    table.insert(globalHistory, shot)
    if #list > 10 then table.remove(list, 1) end
    if #globalHistory > 30 then table.remove(globalHistory, 1) end
    refreshLearnLabel()
end

local function onKick(part, props, speed, now)
    learnKick(part, part.AssemblyLinearVelocity)
    finishShot()
end

```

```

stats.Shots += 1
stats.TotalSpeed += speed
stats.LastSpeed = speed
stats.MaxSpeed = math.max(stats.MaxSpeed, speed)
refreshStats()

table.insert(shotOrigins, part.Position - Vector3.new(0, props.Size.Y / 2, 0))
if #shotOrigins > CONFIG.Analysis.MaxOrigins then
    table.remove(shotOrigins, 1)
end

-- نحفظ التوقع لحظة الركلة (بدون تعويض البندق لأننا نقارن باللي نشوفه)
lastShot = {
    Ball = part,
    StartTime = now,
    Predicted = simulate(part, props, nil, { NoLag = true, Duration = CONFIG.Analysis.RecordTime }),
    Actual = { part.Position },
    ActualTimes = { 0 },
    LastSample = now,
    Done = false,
}
end

local function updateShotRecording(now)
    if not lastShot or lastShot.Done then return end
    local ball = lastShot.Ball
    local elapsed = now - lastShot.StartTime
    if not ball.Parent or elapsed > CONFIG.Analysis.RecordTime then
        finishShot()
    elseif now - lastShot.LastSample >= 1 / 30 then
        lastShot.LastSample = now
        table.insert(lastShot.Actual, ball.Position)
        table.insert(lastShot.ActualTimes, elapsed)
    end
end

local function drawReplay()
    if not lastShot then return end
    local actual = lastShot.Actual
    for i = 1, #actual - 1 do
        drawSegment(actual[i], actual[i + 1], Color3.new(1, 1, 1), 0.15, 0.14)
    end
    local predicted = lastShot.Predicted.Points
    for i = 1, #predicted - 1 do
        drawSegment(predicted[i], predicted[i + 1], outlineColor(), 0.55, 0.12)
    end
end

-- يرسم مسار شوت افتراضي حسب أسلوب اللاعب واتجاهه
local function drawAssistPath(part, props, plr, forward, color, showLabels)
    local shot = learnedShot(plr.UserId)
    if not shot then return false end
    local result = simulate(part, props, nil, {
        Position = part.Position,
        Velocity = fromLocalShot(forward, shot),
    })
    local points = result.Points
    for i = 1, #points - 1 do
        drawSegment(points[i], points[i + 1], color, 0.2 + 0.5 * (i / #points), 0.14)
    end
    local crossing = state.goalEntry and findGoalCrossing(result)
    if crossing then
        drawGoalCrossing(crossing, showLabels)
    end
    return true
end

local function sameTeam(a, b)
    return a.Team ~= nil and a.Team == b.Team
end

local function drawShotAssists(camera)
    if not (state.shotAssist or state.opponentPredict) then return end
    local A = CONFIG.Analysis

    -- مين معه الكرة: أقرب كرة بطيئة لكل لاعب
    local possessions = {}
    for part, props in pairs(trackedBalls) do
        if part.Parent and part.AssemblyLinearVelocity.Magnitude < A.PossessionSpeed then
            for _, plr in ipairs(Players.GetPlayers()) do
                local root = rootOf(plr)

```

```

        if root then
            local distance = (root.Position - part.Position).Magnitude
            local current = possessions[plr]
            if distance < A.PossessionRange and (not current or distance < current.Distance) then
                possessions[plr] = { Part = part, Props = props, Root = root, Distance = distance }
            end
        end
    end
end

local mine = possessions[player]
if state.shotAssist and mine then
    local look = state.assistCamera and camera.CFrame.LookVector or mine.Root.CFrame.LookVector
    local forward = flatForward(look)
    if forward then
        drawAssistPath(mine.Part, mine.Props, player, forward, THEME.Success, state.predictLabels)
    end
end

if state.opponentPredict then
    local count = 0
    for plr, info in pairs(possessions) do
        if count >= A.MaxOpponents then break end
        if plr ~= player and not sameTeam(plr, player) then
            local forward = flatForward(info.Root.CFrame.LookVector)
            if forward and drawAssistPath(info.Part, info.Props, plr, forward, THEME.Danger, false) then
                count += 1
            end
        end
    end
end

local function drawOrigins()
    for _, point in ipairs(shotOrigins) do
        local origin = take("Origin").Part
        origin.CFrame = CFrame.new(point + Vector3.new(0, 0.07, 0)) * CFrame.Angles(0, 0, math.pi / 2)
        origin.Color = THEME.Warning
        origin.Transparency = 0.4
    end
end

local lastInfoUpdate = 0
local lastFilterUpdate = 0
local alertActive = false

connect(RunService.RenderStepped, function()
    local camera = Workspace.CurrentCamera
    local rootPos = getRootPosition()
    local nearest = math.huge
    local now = os.clock()
    local pulse = (math.sin(now * 5) + 1) / 2
    local P = CONFIG.Prediction

    if vizFolder.Parent ~= camera then
        vizFolder.Parent = camera
    end
    if now - lastFilterUpdate > 0.5 then
        lastFilterUpdate = now
        refreshRayFilter()
    end

    local moving = {}

    for part, props in pairs(trackedBalls) do
        if not part.Parent then
            props.Tracer.Visible = false
            continue
        end

        local distance = rootPos and (part.Position - rootPos).Magnitude
        if distance then
            nearest = math.min(nearest, distance)
        end

        -- تطبيق الهيت بوكس
        if state.enabled then
            local mult = state.size
            if state.autoSize and distance then
                mult = math.min(mult * (1 + distance / CONFIG.AutoSizeRange), CONFIG.MaxSize * 1.5)
            end
        end
    end
end)

```

```

end
local targetSize = props.Size * mult * Vector3.new(state.axisX, state.axisY, state.axisZ)
local targetTransparency = state.showHitbox and state.transparency or props.Transparency

if part.Size ~= targetSize then
    part.Size = targetSize
end
if part.Transparency ~= targetTransparency then
    part.Transparency = targetTransparency
end
if part.CanCollide then
    part.CanCollide = false
end
end

-- الإطار + النيص
props.Box.Visible = state.enabled and state.outline
if props.Box.Visible then
    if state.pulse then
        props.Box.LineThickness = 0.03 + 0.06 * pulse
        props.Box.Transparency = 0.5 * (1 - pulse)
    else
        props.Box.LineThickness = 0.05
        props.Box.Transparency = 0
    end
end

end

-- المؤشر: المسافة والسرعة
local velocity = part.AssemblyLinearVelocity
props.Esp.Enabled = state.esp
if state.esp then
    -- المؤشر دائماً فوق الكرة حتى لو الهيتبوكس كبير
    props.Esp.StudsOffset = Vector3.new(0, part.Size.Y / 2 + 1.5, 0)
    props.EspLabel.Text = string.format(
        "🏏 %s 0.% < • م f",
        distance and string.format("%.1f", distance) or "-",
        velocity.Magnitude
    )
end

-- خط التتبع
local ballScreen = camera:WorldToViewportPoint(part.Position)
local rootScreen = rootPos and camera:WorldToViewportPoint(rootPos)
if state.tracer and rootScreen and ballScreen.Z > 0 and rootScreen.Z > 0 then
    props.Tracer.Visible = true
    drawLine(props.Tracer, Vector2.new(rootScreen.X, rootScreen.Y), Vector2.new(ballScreen.X, ballScreen.Y))
else
    props.Tracer.Visible = false
end

end

-- كشف الشوت: زيادة مفاجئة في سرعة الكرة
local speed = velocity.Magnitude
local prevSpeed = props.PrevSpeed or speed
props.PrevSpeed = speed
if speed - prevSpeed > CONFIG.Analysis.KickJump
    and speed > CONFIG.Analysis.MinShotSpeed
    and now - (props.LastKick or 0) > 0.5
then
    props.LastKick = now
    onKick(part, props, speed, now)
end

end

-- المعايرة تشتغل دائماً عشان تكون جاهزة لما تنطلق الكرة
calibrate(part, props, now)

if state.prediction and velocity.Magnitude > P.MinSpeed and (not distance or distance < P.MaxRange) then
    table.insert(moving, { Part = part, Props = props, Distance = distance or 0, Speed = velocity.Magnitude })
end
end

-- توقّع المسار لأقرب الكرات المتحركة
local focus
if state.prediction then
    table.sort(moving, function(a, b)
        return a.Distance < b.Distance
    end)
    local color = outlineColor()
    for i = 1, math.min(#moving, P.MaxBalls) do
        local entry = moving[i]
        local result = simulate(entry.Part, entry.Props, rootPos)
        local part = entry.Part

```

```

-- الكنايات بس للكرة الأقرب عشان ما تزجم الشاشة
local showLabels = state.predictLabels and i == 1
local clearRadius = math.max(part.Size.X, part.Size.Y, part.Size.Z) * 0.6 + 0.5
drawPrediction(result, color, showLabels, part.Position, clearRadius)

local crossing = state.goalEntry and findGoalCrossing(result)
if crossing then
    drawGoalCrossing(crossing, showLabels)
end
if i == 1 then
    focus = { Entry = entry, Result = result, Crossing = crossing }
end
end
end

updateShotRecording(now)
if state.goalEntry and state.showGoalFrame then
    drawGoalFrames()
end
if state.showReplay then
    drawReplay()
end
if state.showOrigins then
    drawOrigins()
end
drawShotAssists(camera)
hideUnused()

-- تنبيه قرب الكرة
if state.alert and nearest <= state.alertDistance then
    if not alertActive then
        alertActive = true
        pcall(function() alertSound:Play() end)
        notify("⚠️ الكرة قريبة منك", THEME.Danger)
    end
elseif nearest > state.alertDistance + 3 then
    alertActive = false
end

-- تحديث المعلومات
if now - lastInfoUpdate > 0.15 then
    lastInfoUpdate = now
    if nearestLabel then
        nearestLabel.Text = nearest < math.huge and string.format("%.1f م", nearest) or "-"
    end
    if predictionLabels.Speed then
        if focus then
            local result = focus.Result
            predictionLabels.Speed.Text = string.format("%.1f", focus.Entry.Speed)
            predictionLabels.Landing.Text = result.Landing and string.format("%.2f ث", result.Landing.Time) or "-"
            predictionLabels.Closest.Text = result.Closest
            and string.format("%.1f م (%.2f ث)", result.Closest.Distance, result.Closest.Time)
            or "-"
            predictionLabels.Calib.Text = string.format("%.1f", focus.Entry.Props.Calib.Magnitude)
            local crossing = focus.Crossing
            predictionLabels.Goal.Text = crossing
            and string.format(crossing.Inside and "ث هدف بعد 2." or "ث برّا (%.2f)", crossing.Time)
            or (#getGoals() == 0 and "حُدّ المرمى من تبويب التحليل" or "ما تروح للمرمى")
        else
            predictionLabels.Speed.Text = "لا توجد كرة متحركة"
            predictionLabels.Landing.Text = "-"
            predictionLabels.Closest.Text = "-"
            predictionLabels.Calib.Text = "-"
            predictionLabels.Goal.Text = "-"
        end
        predictionLabels.Ping.Text = string.format("%d ms", math.floor(networkLag() * 1000 + 0.5))
    end
end
end)

-----
-- صفحة 1: الهيتبوكس
-----
newPage(TAB_NAMES[1])

local function statusSuffix()
    return " • اختصار: " .. state.quickKey
end

local enableCard = card(54)
local statusText = cardTitle(enableCard, "الحالة: متوقف", "تفعيل الهيتبوكس", .. statusSuffix())

```



```

local function refreshStatus()
    statusText.Text = (state.enabled and "الحالة: يعمل ✓" or "الحالة: متوقف") .. statusSuffix()
    statusText.TextColor3 = state.enabled and THEME.Success or THEME.SubText
end

local enableSwitch = makeSwitch(enableCard, state.enabled, function(value)
    state.enabled = value
    refreshStatus()
    if value then
        notify("تم تفعيل الهيتيوكس", THEME.Success)
    else
        restoreAll()
        notify("تم إيقاف الهيتيوكس", THEME.Danger)
    end
end)

-- الحجم + أحجام سريعة
local sizeCard = card(118)
cardHeading(sizeCard, "حجم الهيتيوكس")

local sizeSlider = makeSlider(sizeCard, 48, CONFIG.MinSize, CONFIG.MaxSize, 0.1, state.size, "x%.1f", function(value)
    state.size = value
    scheduleSave()
end)

local resetBtn = smallButton(sizeCard, "إعادة", UDim2.new(0, 80, 0, 10), UDim2.new(0, 56, 0, 24))
resetBtn.MouseButton1Click:Connect(function()
    sizeSlider.set(1, true)
    notify("تمت إعادة الحجم إلى الافتراضي", THEME.Accent2)
end)

local presetRow = horizontalRow(sizeCard, 74, 30)
local presetCount = #CONFIG.SizePresets
for i, preset in ipairs(CONFIG.SizePresets) do
    local btn = smallButton(
        presetRow,
        "x" .. preset,
        UDim2.new(),
        UDim2.new(1 / presetCount, -8 * (presetCount - 1) / presetCount, 1, 0)
    )
    btn.LayoutOrder = i
    btn.TextSize = 13
    btn.MouseButton1Click:Connect(function()
        sizeSlider.set(preset, true)
    end)
end

switchCard("حجم تلقائي حسب المسافة", "يغير الهيتيوكس كلما ابتعدت الكرة", "autoSize")

sectionHeader("تغيير كل اتجاه 🏏")

local axisCard = card(196)
local axisSliders = {}
local axes = {
    { key = "axisX", name = "(X) العرض" },
    { key = "axisY", name = "(Y) الارتفاع" },
    { key = "axisZ", name = "(Z) الطول" },
}
for i, axis in ipairs(axes) do
    local y = 46 + (i - 1) * 60
    cardHeading(axisCard, axis.name, y - 36)
    axisSliders[i] = makeSlider(axisCard, y, 0.25, 3, 0.05, state[axis.key], "x%.2f", function(value)
        state[axis.key] = value
        scheduleSave()
    end)
end

local axisResetCard = card(44)
local axisResetBtn = smallButton(axisResetCard, "إعادة الاتجاهات إلى x1", UDim2.new(0, 14, 0, 8), UDim2.new(1, -28, 0, 28))
axisResetBtn.TextSize = 13
axisResetBtn.MouseButton1Click:Connect(function()
    for _, slider in ipairs(axisSliders) do
        slider.set(1, true)
    end
end)

-----
-- صفحة 2: التوقع
-----
newPage(TAB_NAMES[2])

```

```

switchCard("توقّع مسار الشوت", "prediction", "خط ثلاثي الأبعاد يوضح وين رح تروح الكرة")

local liveCard = card(166)
cardHeading(liveCard, "قراءة مباشرة")
predictionLabels.Speed = statRow(liveCard, 38, "سرعة الكرة")
predictionLabels.Landing = statRow(liveCard, 58, "تنزل بعد")
predictionLabels.Goal = statRow(liveCard, 78, "دخول المرمى")
predictionLabels.Closest = statRow(liveCard, 98, "أقرب مرور منك")
predictionLabels.Calib = statRow(liveCard, 118, "تصحيح المعايير")
predictionLabels.Ping = statRow(liveCard, 138, "البنق")

sliderCard("مدة التوقّع (ثانية)", "predictTime", 0.5, 5, 0.25, "%.2f")

sectionHeader("الدقة")
switchCard("حساب الارتداد", "predictBounces", "يحسب ارتداد الكرة من الأرض والقائم والعارضة")
switchCard("نعويض البنق", "predictLag", "يقدم الكرة لمكانها الحقيقي في السيرفر")
switchCard("معايرة تلقائية", "predictCalibrate", "يتعلم انحناء الكرة من حركتها ويصح المسار")

sectionHeader("العلامات")
switchCard("مكان النزول", "predictLanding", "دائرة مكان نزول الكرة مع الوقت")
switchCard("أقرب نقطة لك", "predictClosest", "وين رح تمر الكرة أقرب شي منك")
switchCard("إظهار الكتابات", "predictLabels", "الوقت والمسافة فوق العلامات")

-----
-- صفحة 3: التحليل
-----

newPage(TAB_NAMES[3])

sectionHeader("المرمى")
switchCard("نقطة دخول المرمى", "goalEntry", "وين ومتى الكرة رح تدخل المرمى")
switchCard("إظهار حدود المرمى", "showGoalFrame", "خطوط توضح المرمى اللي حفظته")

local goalCard = card(124)
cardHeading(goalCard, "حفظ المرمى")
label(goalCard, {
    AnchorPoint = Vector2.new(1, 0),
    Position = UDim2.new(1, -14, 0, 32),
    Size = UDim2.new(1, -28, 0, 30),
    Text = "قف على خط المرمى بالنص، ووجهك للملعب، واضغط حفظ",
    TextColor3 = THEME.SubText,
    TextSize = 12,
    Font = FONT_REG,
    TextWrapped = true,
    TextXAlignment = Enum.TextXAlignment.Right,
})
local goalStatus = label(goalCard, {
    Position = UDim2.new(0, 14, 0, 10),
    Size = UDim2.new(0, 120, 0, 20),
    Text = "",
    TextSize = 12,
    TextXAlignment = Enum.TextXAlignment.Left,
})

local function refreshGoalStatus()
    goalCache = nil
    local n = #getGoals()
    goalStatus.Text = "المحفوظ: " .. n .. " / 2"
    goalStatus.TextColor3 = n > 0 and THEME.Success or THEME.SubText
end

local function saveGoal(index)
    local character = player.Character
    local root = character and character:FindFirstChild("HumanoidRootPart")
    if not root then
        notify("ما لقيت شخصيتك", THEME.Danger)
        return
    end
    local look = root.CFrame.LookVector
    local facing = Vector3.new(look.X, 0, look.Z)
    if facing.Magnitude < 0.1 then return end
    facing = facing.Unit
    local ground = Workspace.Raycast(root.Position, Vector3.new(0, -25, 0), rayParams)
    local base = ground and ground.Position or (root.Position - Vector3.new(0, 3, 0))
    -- يحفظها صج JSON عشان) ما نترك فراغ في القائمة
    index = math.min(index, #state.goals + 1)
    state.goals[index] = { base.X, base.Y, base.Z, facing.X, facing.Z }
    scheduleSave()
    refreshGoalStatus()
    notify("تم حفظ المرمى " .. index .. " ✓", THEME.Success)
end

```

```

local goalRow = horizontalRow(goalCard, 80, 30)
for i, info in ipairs({
    { "حفظ مرمى 1", function() saveGoal(1) end },
    { "حفظ مرمى 2", function() saveGoal(2) end },
    { "مسح", function()
        state.goals = {}
        scheduleSave()
        refreshGoalStatus()
        notify("تم مسح المرامي", THEME.Accent2)
    end },
}) do
    local btn = smallButton(goalRow, info[1], UDim2.new(), UDim2.new(1 / 3, -6, 1, 0))
    btn.LayoutOrder = i
    btn.MouseButton1Click:Connect(info[2])
end
refreshGoalStatus()

sliderCard("عرض المرمى", "goalWidth", 6, 50, 0.5, "%.1f")
sliderCard("ارتفاع المرمى", "goalHeight", 3, 20, 0.5, "%.1f")

sectionHeader("إحصائيات الشوتات 📊")
local statsCard = card(156)
statLabels.Shots = statRow(statsCard, 12, "عدد الشوتات")
statLabels.Max = statRow(statsCard, 34, "أقصى شوت")
statLabels.Avg = statRow(statsCard, 56, "متوسط السرعة")
statLabels.Last = statRow(statsCard, 78, "آخر شوت")
statLabels.Error = statRow(statsCard, 100, "خطأ التوقع (متوسط)")
local resetStatsBtn = smallButton(statsCard, "تفسير الإحصائيات", UDim2.new(0, 14, 0, 122), UDim2.new(1, -28, 0, 26))
resetStatsBtn.MouseButton1Click:Connect(function()
    stats = { Shots = 0, MaxSpeed = 0, TotalSpeed = 0, LastSpeed = 0, LastError = nil }
    table.clear(shotOrigins)
    lastShot = nil
    refreshStats()
end)
refreshStats()

sectionHeader("آخر شوت 📺")
switchCard("عرض آخر شوت", "الحقيقي (أبيض جنب المتوقع (ملون)", "showReplay")
switchCard("علامة مكان آخر 20 شوت", "أماكن التسديد", "showOrigins")

sectionHeader("مساعد التسديد 🧠")
switchCard("خط شوتك المتوقع", "لما الكرة عندك: وين رح تروح إذا سددت (أحضر)", "shotAssist")
switchCard("حسب اتجاه الكاميرا", "شغله إذا الشوت في الماب يمشي مع الكاميرا", "assistCamera")
switchCard("توقع شوت الخصم", "تقريبي: من اتجاهه وأسلوب شواته (أحمر)", "opponentPredict")

local learnCard = card(58)
label(learnCard, {
    AnchorPoint = Vector2.new(1, 0),
    Position = UDim2.new(1, -14, 0, 8),
    Size = UDim2.new(1, -28, 0, 18),
    Text = "يتعلم من كل شوت ينسدد في السيرفر",
    TextColor3 = THEME.SubText,
    TextSize = 12,
    Font = FONT_REG,
    TextXAlignment = Enum.TextXAlignment.Right,
})
learnLabel = label(learnCard, {
    AnchorPoint = Vector2.new(1, 0),
    Position = UDim2.new(1, -14, 0, 30),
    Size = UDim2.new(1, -28, 0, 18),
    Text = "",
    TextSize = 13,
    TextXAlignment = Enum.TextXAlignment.Right,
})
refreshLearnLabel()

-----
-- صفحة 4: المظهر
-----

newPage(TAB_NAMES[4])

switchCard("إظهار الهيت بوكس", "جعل الكرة شفافة لرؤية الحجم", "showHitbox")
sliderCard("شفافية الكرة", "transparency", 0, 0.95, 0.05, "%.2f")
switchCard("إطار الهيت بوكس", "خطوط ملونة توضح حدود الكرة", "outline")
switchCard("تأثير النبض", "الإطار يلمع ويتحرك باستمرار", "pulse")

local colorCard = card(84)
cardHeading(colorCard, "لون الإطار والمسار")

local colorRow = horizontalRow(colorCard, 42, 30, Enum.HorizontalAlignment.Right)

```

```

local colorStrokes = {}

local function selectColor(index)
    state.colorIndex = index
    scheduleSave()
    for i, s in ipairs(colorStrokes) do
        tween(s, 0.2, { Transparency = (i == index) and 0 or 1 })
    end
    local color = outlineColor()
    for _, props in pairs(trackedBalls) do
        props.Box.Color3 = color
        props.Tracer.BackgroundColor3 = color
    end
end

for i, color in ipairs(CONFIG.OutlineColors) do
    local swatch = create("TextButton", {
        Size = UDim2.new(0, 30, 0, 30),
        BackgroundColor3 = color,
        AutoButtonColor = false,
        Text = "",
        LayoutOrder = i,
        Parent = colorRow,
    }, { corner(15) })
    local s = stroke(THEME.Text, 2, 1)
    s.Parent = swatch
    colorStrokes[i] = s
    swatch.MouseButton1Click:Connect(function()
        selectColor(i)
    end)
end

local themeCard = card(84)
cardHeading(themeCard, "نيم الواجهة")

local themeRow = horizontalRow(themeCard, 42, 30, Enum.HorizontalAlignment.Right)
local themeStrokes = {}

local function applyTheme(index)
    state.themeIndex = index
    scheduleSave()
    local theme = CONFIG.Themes[index]
    THEME.Accent = theme.Accent
    THEME.Accent2 = theme.Accent2
    local sequence = ColorSequence.new(THEME.Accent, THEME.Accent2)
    for _, g in ipairs(gradients) do
        g.Color = sequence
    end
    for _, ref in ipairs(accentRefs) do
        ref[1][ref[2]] = THEME[ref[3]]
    end
    for i, s in ipairs(themeStrokes) do
        tween(s, 0.2, { Transparency = (i == index) and 0 or 1 })
    end
end

for i, theme in ipairs(CONFIG.Themes) do
    local btn = create("TextButton", {
        Size = UDim2.new(0, 64, 0, 30),
        BackgroundColor3 = Color3.new(1, 1, 1),
        AutoButtonColor = false,
        Text = theme.Name,
        TextColor3 = Color3.new(1, 1, 1),
        TextStrokeTransparency = 0.6,
        TextSize = 12,
        Font = FONT_BOLD,
        LayoutOrder = i,
        Parent = themeRow,
    }, {
        corner(8),
        create("UIGradient", { Color = ColorSequence.new(theme.Accent, theme.Accent2) }),
    })
    local s = stroke(THEME.Text, 2, 1)
    s.Parent = btn
    themeStrokes[i] = s
    btn.MouseButton1Click:Connect(function()
        applyTheme(i)
        notify("تم تغيير النيم إلى" .. theme.Name, THEME.Accent)
    end)
end

```

```

-----
-- صفحة 5: أدوات --
-----

newPage(TAB_NAMES[5])

switchCard("مؤشر الكرة", "esp", "المسافة والسرعة فوق الكرة من خلف الجدران")
switchCard("خط التتبع", "tracer", "خط من شخمينك إلى الكرة")
switchCard("تنبيه قرب الكرة", "alert", "صوت وإشعار لما تقترب الكرة منك")
sliderCard("مسافة التنبيه", "alertDistance", 5, 50, 1, "%d م")

local nameCard = card(84)
cardHeading(nameCard, "اسم الكرة في اللعبة")

local nameBox = create("TextBox", {
    AnchorPoint = Vector2.new(1, 0),
    Position = UDim2.new(1, -14, 0, 40),
    Size = UDim2.new(1, -100, 0, 30),
    BackgroundColor3 = THEME.SurfaceLight,
    Text = state.ballName,
    PlaceholderText = "مثال: Ball",
    PlaceholderColor3 = THEME.SubText,
    TextColor3 = THEME.Text,
    TextSize = 14,
    Font = FONT_BOLD,
    ClearTextOnFocus = false,
    Parent = nameCard,
}, { corner(6), stroke(THEME.Stroke, 1) })

local searchBtn = smallButton(nameCard, "بحث", UDim2.new(0, 14, 0, 40), UDim2.new(0, 66, 0, 30))
searchBtn.TextSize = 13

local function applyBallName()
    local name = nameBox.Text:gsub("^s+", ""):gsub("%s+$", "")
    if name == "" then
        nameBox.Text = state.ballName
        return
    end
    state.ballName = name
    scheduleSave()
    rescan()
    local n = countBalls()
    if n > 0 then
        notify("كرة باسم " .. name, THEME.Success)
    else
        notify("لا توجد كرة باسم", THEME.Danger)
    end
end

searchBtn.MouseButton1Click:Connect(applyBallName)
nameBox.FocusLost:Connect(function(enterPressed)
    if enterPressed then
        applyBallName()
    end
end)

```

```

-----
-- صفحة 6: الإعدادات --
-----

newPage(TAB_NAMES[6])

-- تغيير أزرار الاختصار
local capturing = nil -- { key = "toggleKey", button = btn }
local keyInfoLabel -- يُنشأ في بطاقة المعلومات

local function refreshKeyInfo()
    if keyInfoLabel then
        keyInfoLabel.Text = state.toggleKey .. " / " .. state.quickKey
    end
    refreshStatus()
end

local function keybindCard(title, key)
    local c = card(54)
    cardTitle(c, title, "اضغط على الزر ثم اختر مفتاحاً جديداً")
    local btn = smallButton(c, state[key], UDim2.new(0, 14, 0, 12), UDim2.new(0, 92, 0, 30))
    btn.TextSize = 13
    btn.MouseButton1Click:Connect(function()
        if capturing then
            capturing.button.Text = state[capturing.key]
        end
        capturing = { key = key, button = btn }
    end)
end

```

```

        btn.Text = "اضغط زر..."
    end)
end

keybindCard("زر إظهار الواجهة", "toggleKey")
keybindCard("زر تشغيل الهيتيوكس", "quickKey")

-- الزر العائم (للجوال)
local floatingBtn = create("TextButton", {
    AnchorPoint = Vector2.new(0.5, 0.5),
    Position = UDim2.new(0, 50, 0.5, 0),
    Size = UDim2.new(0, 52, 0, 52),
    BackgroundColor3 = Color3.new(1, 1, 1),
    AutoButtonColor = false,
    Text = "🕸",
    TextSize = 24,
    Font = FONT_BOLD,
    TextColor3 = THEME.Text,
    Visible = state.floatingButton,
    ZIndex = 20,
    Parent = screenGui,
}, { corner(26), accentGradient(45), stroke(THEME.Background, 2) })

switchCard("زر عائم", "floatingButton", function(value)
    floatingBtn.Visible = value
end)

-- حفظ الإعدادات
local saveCard = card(54)
cardTitle(
    saveCard,
    "حفظ الإعدادات",
    canSave and "✓ يتم الحفظ تلقائياً" or "لا يدعم الحفظ Executor"
)
local resetSettingsBtn = smallButton(saveCard, "حذف المحفوظ", UDim2.new(0, 14, 0, 12), UDim2.new(0, 92, 0, 30))
resetSettingsBtn.MouseButton1Click:Connect(function()
    if not canSave then
        notify("Executor الحفظ غير مدعوم في هذا الـ", THEME.Danger)
        return
    end
    pcall(function()
        if typeof(delfile) == "function" then
            delfile(CONFIG.SaveFile)
        else
            writefile(CONFIG.SaveFile, "{}")
        end
    end)
    notify("تم حذف الإعدادات المحفوظة - أعد تشغيل السكريبت", THEME.Accent2)
end)

sectionHeader("معلومات")

local infoCard = card(86)
ballCountLabel = statRow(infoCard, 12, "الكرات المكتشفة")
nearestLabel = statRow(infoCard, 34, "أقرب كرة")
keyInfoLabel = statRow(infoCard, 56, "الواجهة / الهيتيوكس")

sectionHeader("عن السكريبت")

local links = {}
for _, link in ipairs(CONFIG.Links) do
    if link.Url ~= "" then
        table.insert(links, link)
    end
end

local aboutCard = card(96 + #links * 38)

local aboutTitle = label(aboutCard, {
    Position = UDim2.new(0, 14, 0, 12),
    Size = UDim2.new(1, -28, 0, 24),
    Text = "• " .. CONFIG.Author .. " •",
    TextColor3 = Color3.new(1, 1, 1),
    TextSize = 20,
})
accentGradient(0).Parent = aboutTitle

label(aboutCard, {
    Position = UDim2.new(0, 14, 0, 40),
    Size = UDim2.new(1, -28, 0, 16),
    Text = "• " .. CONFIG.Version,

```

```

        TextColor3 = THEME.SubText,
        TextSize = 12,
        Font = FONT_REG,
    })

    label(aboutCard, {
        Position = UDim2.new(0, 14, 0, 62),
        Size = UDim2.new(1, -28, 0, 18),
        Text = "شكراً لاستخدامك السكرت",
        TextSize = 13,
    })

    for i, link in ipairs(links) do
        local y = 92 + (i - 1) * 38
        label(aboutCard, {
            AnchorPoint = Vector2.new(1, 0),
            Position = UDim2.new(1, -14, 0, y),
            Size = UDim2.new(1, -120, 0, 30),
            Text = link.Name,
            TextSize = 13,
            TextXAlignment = Enum.TextXAlignment.Right,
        })
        local copyBtn = smallButton(aboutCard, "نسخ الرابط", UDim2.new(0, 14, 0, y), UDim2.new(0, 92, 0, 30))
        copyBtn.MouseButton1Click:Connect(function()
            if typeof(setclipboard) == "function" then
                pcall(setclipboard, link.Url)
                notify("تم نسخ رابط " .. link.Name, THEME.Success)
            else
                notify(link.Url, THEME.Accent2)
            end
        end)
    end

    -----
    -- السحب والإدخال
    -----

    local windowDragging = false
    local dragStart, startPos

    local floatPressed = false
    local floatMoved = false
    local floatStart, floatStartPos

    titleBar.InputBegan:Connect(function(input)
        if isPress(input) then
            windowDragging = true
            dragStart = input.Position
            startPos = mainFrame.Position
        end
    end)

    floatingBtn.InputBegan:Connect(function(input)
        if isPress(input) then
            floatPressed = true
            floatMoved = false
            floatStart = input.Position
            floatStartPos = floatingBtn.Position
        end
    end)

    connect(UserInputService.InputChanged, function(input)
        if not isMove(input) then return end

        if activeSlider then
            activeSlider.update(input.Position.X)
        elseif windowDragging then
            local delta = input.Position - dragStart
            mainFrame.Position = UDim2.new(
                startPos.X.Scale, startPos.X.Offset + delta.X,
                startPos.Y.Scale, startPos.Y.Offset + delta.Y
            )
        elseif floatPressed then
            local delta = input.Position - floatStart
            if delta.Magnitude > 6 then
                floatMoved = true
            end
            if floatMoved then
                floatingBtn.Position = UDim2.new(
                    floatStartPos.X.Scale, floatStartPos.X.Offset + delta.X,
                    floatStartPos.Y.Scale, floatStartPos.Y.Offset + delta.Y
                )
            )
        end
    end)

```

```

        end
    end
end)

-- إظهار / إخفاء
local function setVisible(visible)
    state.guiVisible = visible
    if visible then
        mainFrame.Visible = true
        tween(mainScale, 0.35, { Scale = 1 }, Enum.EasingStyle.Back)
    else
        tween(mainScale, 0.25, { Scale = 0 }).Completed:Connect(function()
            if not state.guiVisible then
                mainFrame.Visible = false
            end
        end)
    end
end

end

end

connect(UserInputService.InputEnded, function(input)
    if isPress(input) then
        if activeSlider then
            activeSlider.release()
            activeSlider = nil
        end
        if floatPressed and not floatMoved then
            setVisible(not state.guiVisible)
        end
        floatPressed = false
        windowDragging = false
    end
end)

connect(UserInputService.InputBegan, function(input, gameProcessed)
    -- التقاط زر اختصار جديد
    if capturing and input.UserInputType == Enum.UserInputType.Keyboard then
        local c = capturing
        capturing = nil
        if input.KeyCode ~= Enum.KeyCode.Escape then
            state[c.key] = input.KeyCode.Name
            scheduleSave()
            notify("تم تعيين الزر: " .. input.KeyCode.Name, THEME.Success)
        end
        c.button.Text = state[c.key]
        refreshKeyInfo()
        return
    end

    if gameProcessed then return end
    if input.KeyCode.Name == state.toggleKey then
        setVisible(not state.guiVisible)
    elseif input.KeyCode.Name == state.quickKey then
        enableSwitch.set(not state.enabled)
    end
end)

-- التصغير
minimizeBtn.MouseButton1Click:Connect(function()
    state.minimized = not state.minimized
    minimizeBtn.Text = state.minimized and "+" or "-"
    tween(mainFrame, 0.35, { Size = state.minimized and MINI_SIZE or FULL_SIZE })
end)

-- إغلاق الكامل
local function cleanup()
    state.enabled = false
    for part in pairs(trackedBalls) do
        untrackBall(part)
    end
    for _, c in ipairs(connections) do
        c:Disconnect()
    end
    table.clear(connections)
    if vizFolder then
        vizFolder:Destroy()
    end
    if screenGui then
        screenGui:Destroy()
    end
    _G.MohammedTN_Hitbox_Cleanup = nil
end
end

```



```
_G.MohammedTN_Hitbox_Cleanup = cleanup

closeBtn.MouseButton1Click:Connect(function()
    tween(mainScale, 0.25, { Scale = 0 }).Completed:Wait()
    cleanup()
end)

-----
-- البدء
-----

selectColor(state.colorIndex)
applyTheme(state.themeIndex)
selectTab(math.clamp(math.floor(state.activeTab), 1, TAB_COUNT), false)
scanForBalls()
refreshBallCount()
refreshRayFilter()
refreshKeyInfo()
setVisible(true)
notify("صنع من قبل - " .. CONFIG.Version .. " مرحباً! تم تحميل السكريبت" .. CONFIG.Author, THEME.Accent)
```